

Miscellaneous/Review

Yin

1. (**August 2021 Problem 1**) On this problem, only the answer will be graded.

- (a) Put $R = \mathbb{Z}/12\mathbb{Z}$. For which elements $f \in R$ is the localization $R_f = 0$?
- (b) Put $R = \mathbb{Z}/12\mathbb{Z}$. For which elements $f \in R$ is the localization R_f a field? (Recall that the zero ring is not a field.)
- (c) The Jordan form of an operator $A : \mathbb{C}^6 \rightarrow \mathbb{C}^6$ is

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

What is the Jordan form of the operator $A^3 - A$?

2. (**January 2022 Problem 1**) On this problem, only the answer will be graded. Consider the symmetric group S_4 .

- (a) What is the order of S_4 ?
- (b) What is the order of its center $Z(S_4)$?
- (c) What is the order of its commutator subgroup $[S_4, S_4]$?
- (d) Is S_4 a simple group?
- (e) Is S_4 a solvable group?
- (f) How many conjugacy classes does S_4 have?

3. (**August 2022 Problem 1**) On this problem, only your answer will be graded, no justification is necessary.

- (a) Let $K = \mathbb{Q}(2^{1/3})$. Is $K/\mathbb{Q}$ a Galois extension?
- (b) What is the radical of the ideal $(x^3 - x^2) \subset \mathbb{R}[x]$?
- (c) How many different (up to isomorphism) groups of order 49 are there?
- (d) Give an example of two non-zero abelian groups M, N such that $M \otimes_{\mathbb{Z}} N = 0$.
- (e) How many two-sided ideals are there in $M_2(\mathbb{C})$, the ring of 2×2 matrices with entries in $\mathbb{C}$?

Yang

4. (August 2021 Problem 5) Let A be an algebra over $\mathbb{C}$ with two generators e and f and defining relations $e^2 = e, f^2 = f$.

- (a) Let B be the ring of 2×2 matrices over the polynomial ring $\mathbb{C}[t]$. Consider B as an (infinite-dimensional) algebra over $\mathbb{C}$. Show that there exists a unique $\mathbb{C}$ -algebra homomorphism $\phi : A \rightarrow B$ such that $\phi(e) = \begin{pmatrix} 1 & t \\ 0 & 0 \end{pmatrix}$ and $\phi(f) = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$.
- (b) Prove that the homomorphism ϕ from part (a) is injective.
- (c) Consider the algebra A as a vector space over $\mathbb{C}$. Write down a basis for A and prove that your answer is correct.

5. (August 2016 Problem 4) Let R be the subring of $\mathbb{C}[x, y]$ consisting of the polynomials $P(x, y)$ such that $P(x, y) = P(y, x)$.

- (a) Show that R is generated as a $\mathbb{C}$ -algebra by $x + y$ and xy .
- (b) Show that the map $\mathbb{C}[u, v] \rightarrow R$ sending u to $x + y$ and v to xy is an isomorphism.
- (c) Let S be the subring of $\mathbb{C}[x, y]$ consisting of the polynomials $P(x, y)$ such that $P(x, y) = P(-x, -y)$. Can S be generated as a $\mathbb{C}$ -algebra by two polynomials? Either give two generators, or prove that S is not generated by 2 polynomials.

6. (August 2014 Problem 4) Let R be a commutative ring and M an R -module. Recall that a prime ideal P is an associated prime of M if there exists some $m \in M$ such that P consists of those f such that $fm = 0$; i.e. P is the annihilator of some $m \in M$.

- (a) Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be an exact sequence of R -modules, and let P be an associated prime of M . Prove that P is an associated prime of either M' or M'' .
- (b) Is the converse true? That is, if P is an associated prime of either M' or M'' , must it be the case that P is an associated prime of M ?

7. (August 2019 Problem 3) Let L/K be a Galois extension of fields with Galois group G . Let S be the subset of roots of unity in L ; that is, S is the set of elements $x \in L$ such that $x^k = 1$ for some positive integer k .

- (a) Show that the action of G on L restricts to an action of G on S (i.e., the action of G preserves S .)
- (b) Let $L = \mathbb{Q}(i)$ and $K = \mathbb{Q}$. Show in this case that G acts faithfully on S .
- (c) Give an example of L/K such that the action of G on S is not faithful.
- (d) Suppose that the action of G on S is transitive. Prove that $\text{char}(K) = 2$.